Aggregating RCTA Individual Forecasts

Andy Hou

Memo: Aggregating RCTA Individual Forecasts

Executive Summary

This memo analyzes individual forecasts from the first-year HFC RCTA competition. The statistical problem is to construct, for each question-day, a crowd probability vector from individual forecasts and to compare aggregation rules by Brier loss. I reconstruct five baseline aggregation methods from `rct-a-prediction-sets.csv`, score them against outcomes from `rct-a-questions-answers.csv`, and then propose a sixth method: **site-balanced bounded evidence pooling**. The method clips individual probabilities to `[0.02, 0.98]`, pools clipped log-odds within forecasting site, and then combines sites with equal weight. It improves over all five baselines under both scoring summaries: question-balanced Brier decreases from 0.235399 for the median baseline to 0.228379, and question-day weighted Brier decreases from 0.218701 for geometric mean to 0.211749. The paired bootstrap improvement over median is 0.0070198 with 95% interval `[-0.0003299, 0.0137224]`, so the effect is favorable but modest. In practical terms, the method limits how much any one forecaster can move the aggregate and treats each forecasting platform as one information source rather than counting all of its members as independent evidence.

1. Data and Unit of Analysis

The raw RCTA data consist of three files. The questions-answers file contains question metadata, answer options, dates, and resolved outcomes. The prediction-sets file contains individual-level forecasts; a prediction set is one forecaster's submission for one question, with multiple rows for multi-option questions. The daily-forecasts file contains performer-method forecasts rather than individual forecasts, so it is not used as the source for aggregation.

The target unit is a question-day. Let $q \in \{1, \dots, Q\}$ index questions, $t \in \mathcal{T}_q$ index scoring days for question q, $k \in \{1, \dots, K_q\}$ index answer options, and $i \in \mathcal{I}_{qt}$ index forecasters active for question-day (q, t). Let $s(i)$ denote the forecasting site of forecaster i. I write

$$\mathcal{S}_{qt} = \{s(i) : i \in \mathcal{I}_{qt}\}, \qquad n_{sqt} = |\{i \in \mathcal{I}_{qt} : s(i) = s\}|$$

for the set of sites and the number of forecasters from site s on question-day (q, t). The individual forecast vector is

$$p_{iqt} = (p_{iqt1}, \dots, p_{iqtK_q}), \qquad \sum_{k=1}^{K_q} p_{iqtk} = 1,$$

and the resolved outcome vector is

$$y_q = (y_{q1}, \dots, y_{qK_q}), \qquad y_{qk} \in \{0, 1\}, \qquad \sum_{k=1}^{K_q} y_{qk} = 1.$$

For the 166 one-row binary questions, the stored answer option is interpreted as the focal Yes answer. I construct

$$p_{iqt,\mathrm{No}} = 1 - p_{iqt,\mathrm{Yes}}, \qquad y_{q,\mathrm{No}} = 1 - y_{q,\mathrm{Yes}}.$$

The cleaned scored sample contains 189 questions and 13,129 question-days.

2. Cleaning and Scoring-Day Construction

Forecast rows were retained only if they had non-missing and valid forecast probabilities, non-missing resolved probabilities, valid question and prediction-set identifiers, valid answer identifiers, non-missing forecaster identifiers, non-missing timestamps, and were not marked as made after correctness was known. Timestamps were converted to America/New_York. The scoring day is defined by the HFC 2:01pm ET cutoff: forecasts before 2:01pm ET are assigned to that calendar day, while forecasts at or after 2:01pm ET are assigned to the next scoring day.

For each (q, t, i), I selected the latest prediction set by timestamp and retained all answer rows from that prediction set. This preserves the vector-valued nature of multi-option submissions. Selecting answer rows independently would create artificial forecast vectors that no forecaster actually submitted.

3. Loss Functions and Estimands

For binary and nominal questions, the Brier loss for an aggregate forecast $\hat{p}_{qt}$ is

$$L_{qt}(\hat{p}) = \sum_{k=1}^{K_q} (\hat{p}_{qtk} - y_{qk})^2.$$

For binary questions this is the original 0-2 Brier scale:

$$L_{qt}(\hat{p}) = (\hat{p}_{\mathrm{Yes}} - y_{\mathrm{Yes}})^2 + (\hat{p}_{\mathrm{No}} - y_{\mathrm{No}})^2 = 2(\hat{p}_{\mathrm{Yes}} - y_{\mathrm{Yes}})^2.$$

For ordinal questions, I use the Jose-Nau-Winkler ordered Brier score. Let

$$F_j = \sum_{k \le j} \hat{p}_{qtk}, \qquad O_j = \sum_{k \le j} y_{qk}, \qquad j = 1, \dots, K_q - 1.$$

The ordered score is

$$L_{qt}^{\mathrm{ord}}(\hat{p}) = \frac{2}{K_q - 1} \sum_{j=1}^{K_q - 1} (F_j - O_j)^2.$$

I report two empirical risk summaries. The question-day weighted risk is

$$\widehat{R}_{\mathrm{QD}}(m) = \frac{1}{\sum_q |\mathcal{T}_q|} \sum_{q=1}^{Q} \sum_{t\in\mathcal{T}_q} L_{qt}(\hat{p}_{qt}^{(m)}),$$

where each scored question-day receives one vote. The question-balanced risk is

$$\widehat{R}_{\mathrm{QB}}(m) = \frac{1}{Q} \sum_{q=1}^{Q} \left[\frac{1}{|\mathcal{T}_q|} \sum_{t\in\mathcal{T}_q} L_{qt}(\hat{p}_{qt}^{(m)}) \right],$$

where each question receives one vote regardless of how long it remained open. The local HFC scoring document clearly establishes daily Brier / MDB, but does not fully specify the second-stage aggregation for this work-test comparison. I therefore report both estimands.

4. Baseline Aggregation Operators

For each question-day and answer option, I compute five baseline aggregation rules. Let p_{ik} abbreviate p_{iqtk} within a fixed (q, t). The raw mean is

$$a_k = \frac{1}{n}\sum_{i=1}^{n} p_{ik}, \qquad \hat{p}_k = \frac{a_k}{\sum_j a_j}.$$

The median is

$$a_k = \mathrm{median}_i(p_{ik}), \qquad \hat{p}_k = \frac{a_k}{\sum_j a_j}.$$

The geometric mean of probabilities is

$$a_k = \exp\left(\frac{1}{n}\sum_{i=1}^{n} \log(\mathrm{clip}(p_{ik}, \epsilon, 1-\epsilon))\right), \qquad \hat{p}_k = \frac{a_k}{\sum_j a_j}.$$

The trimmed mean removes the top and bottom 10% of scalar probabilities before averaging. The geometric mean of odds is implemented as one-vs-rest log-odds pooling:

$$z_k = \frac{1}{n}\sum_{i=1}^{n} \log\left[\frac{\mathrm{clip}(p_{ik}, \epsilon, 1-\epsilon)}{1-\mathrm{clip}(p_{ik}, \epsilon, 1-\epsilon)}\right], \qquad a_k = \sigma(z_k), \qquad \hat{p}_k = \frac{a_k}{\sum_j a_j}.$$

For binary questions, this reduces to

$$\hat{p}_{\mathrm{Yes}} = \sigma\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{logit}(p_{i,\mathrm{Yes}})\right), \qquad \hat{p}_{\mathrm{No}} = 1 - \hat{p}_{\mathrm{Yes}}.$$

The implementation uses $\epsilon = 10^{-6}$ for the original geometric and odds-based baselines.

5. Baseline Results

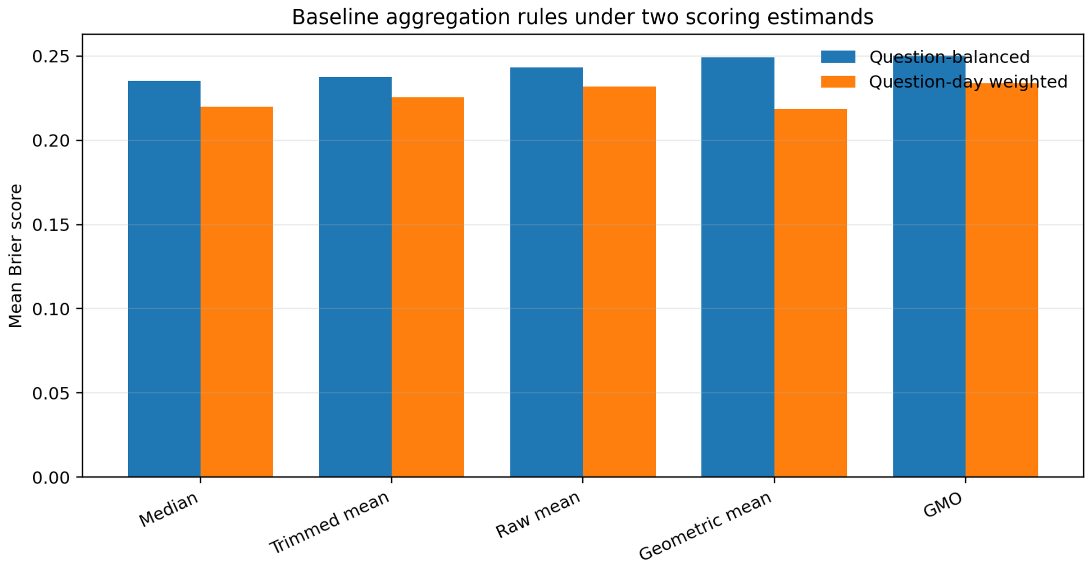

Figure 1: Baseline scores under two estimands

Method	Question-Balanced Brier	Question-Day Weighted Brier
Median	0.235399	0.219796
Trimmed mean	0.237679	0.225603
Raw mean	0.243280	0.231857
Geometric mean	0.249192	0.218701
Geometric mean of odds	0.250367	0.233909

The two estimands give different rankings. Under $\widehat{R}_{\mathrm{QD}}$, geometric mean is best. Under $\widehat{R}_{\mathrm{QB}}$, median is best. This reversal indicates a robustness tradeoff. Geometric methods perform well on many scored days, but they can incur large losses on a smaller number of questions. Median is more stable at the question level, but it does not accumulate evidence on a log scale.

6. Diagnostic Analysis

Before proposing an improved aggregation rule, I diagnosed which failure modes are supported by the data. The key diagnostic was the Murphy decomposition, written on the standard 0-1 Brier scale as

$$\mathrm{BS} = \mathrm{Reliability} - \mathrm{Resolution} + \mathrm{Uncertainty}.$$

Reliability measures calibration error; resolution measures the ability to separate events from non-events. For median, the reliability share is 0.008. For GMO, it is 0.026. Thus the leading baselines are not dominated by calibration error.

Diagnostic	Empirical result	Implication
Murphy decomposition	median rel_share = 0.008; GMO rel_share = 0.026	Global recalibration / extremization has little target
Type stratification	binary 0.147, nominal 0.487, ordinal 0.253 under median question-day weighted scoring	Question geometry matters
Dispersion stratification	low dispersion 0.141, high dispersion 0.273	Disagreement marks difficult question-days
Forecast age	same-day 0.219, one-day 0.220	Much weaker than type and dispersion in this no-carry-forward pipeline
Staleness interpretation	retained forecasts are same-day or one-day latest submissions, not carried-forward stale forecasts	The pipeline limits what recency decay can change

These diagnostics rule out the most standard improvement in the forecasting literature: global extremization. Extremization is useful when an average forecast is too close to 0.5 and should be pushed outward. Here, the leading baselines are not mainly underconfident. The problem is on the other side. Evidence-scale pooling is often useful, but unbounded log-odds evidence can become too extreme.

This gives the final method a clear relation to the literature. Extremization corrects under-accumulated evidence by moving an average forecast away from 0.5. Bounded evidence pooling corrects over-accumulated evidence by limiting how far one forecaster can move the log-odds average. They are mirror-image operations on the same calibration spectrum: one pushes a conservative aggregate outward, while the other pulls an overconfident evidence pool back inward.

7. Tested Conditional Improvements

I also tested the feature classes suggested by the prompt. The results are summarized in Table 4.

Candidate direction	Statistical object	Result	Decision
Skill weighting	Prior resolved questions per active forecaster	First-half median = 3.5; second-half median = 16	Not stable enough for season-wide use
Type-conditioned selection	Walk-forward best method by question type	Oracle improves slightly; walk-forward underperforms median	Hindsight structure is not deployable signal
Thin-crowd override	Switch method when n_{qt} is small	Best threshold mostly collapses to median	Not a meaningful improvement
Dispersion shrinkage	Shrink high-dispersion forecasts toward uniform	Brier worsens monotonically as shrinkage increases	Dispersion is a marker, not an intervention
Activity/update history	Accuracy change after forecast updates	Large updates improve 79% of the time	Interesting future signal, not final rule

The negative results are important. They show that the final method is not a default fallback; it is chosen after plausible leakage-free alternatives fail for identifiable statistical reasons.

8. GMO Tail Autopsy

Let

$$\Delta_{qt} = L_{qt}(\hat{p}^{\mathrm{GMO}}) - L_{qt}(\hat{p}^{\mathrm{median}}).$$

Negative Δ_{qt} means GMO beats median on that question-day. GMO beats median on 68.0% of all question-days and 75.3% of binary question-days. However, positive excess loss is concentrated in a small number of questions. In binary questions, the worst 10 questions account for 85.2% of GMO's positive excess loss relative to median, and the worst 20 account for 97.0%.

This is the empirical basis for the final method. The issue is not that log-odds pooling is always poor. Rather, it is a high-variance evidence-pooling rule: often useful, occasionally catastrophic. Since the original GMO clips only at 10^{-6}, one near-certain forecast can contribute approximately ± 13.8 log-odds units. That is an implausibly large individual evidence contribution in geopolitical forecasting.

The relative-to-individual diagnostic gives the positive side of the same story. GMO's mean percentile rank among individual forecasters is 0.590 on binary question-days and 0.600 on ordinal question-days, compared with 0.447 and 0.563 for median. So GMO often extracts crowd signal better than the median. Its problem is not typical performance; it is tail risk.

9. Final Method: Site-Balanced Bounded Evidence Pooling

The final method modifies GMO in two ways. First, it bounds individual evidence. Second, it changes the aggregation unit from forecaster to site.

In plain language, I first limit how much any one forecaster can influence the aggregate by capping extreme probabilities. I then average forecasts within each forecasting site and combine sites with equal weight. This prevents one overconfident forecaster, or one large site, from dominating the crowd forecast.

The method satisfies the prompt's temporal constraint directly. It uses the forecasts available on day T, time-invariant site labels, and a fixed cap $c = 0.02$. The cap is not fit from resolved outcomes and does not use information from day T or later.

For binary questions, set $c = 0.02$ and define

$$p_i^{(c)} = \min\{\max(p_i, c), 1 - c\}.$$

The individual bounded evidence is

$$z_i^{(c)} = \mathrm{logit}(p_i^{(c)}).$$

Within site s, pool evidence by

$$\bar{z}_s = \frac{1}{n_s} \sum_{i:s(i)=s} z_i^{(c)}.$$

Across sites, combine site-level evidence by

$$\bar{z} = \frac{1}{|\mathcal{S}_{qt}|} \sum_{s \in \mathcal{S}_{qt}} \bar{z}_s, \qquad \hat{p}_{\text{Yes}} = \sigma(\bar{z}).$$

For multi-option questions, I use the one-vs-rest extension. For each option k,

$$\bar{z}_{sk} = \frac{1}{n_s} \sum_{i:s(i)=s} \operatorname{logit}(p_{ik}^{(c)}), \qquad \bar{z}_k = \frac{1}{|\mathcal{S}_{qt}|} \sum_s \bar{z}_{sk}, \qquad a_k = \sigma(\bar{z}_k), \qquad \hat{p}_k = \frac{a_k}{\sum_j a_j}.$$

The cap $c = 0.02$ implies that no individual forecast contributes more than $49:1$ odds. This is a modeling assumption, not merely a numerical safeguard. It says that individual near-certainty is treated as bounded evidence rather than an unbounded likelihood ratio.

The following sensitivity check varies the cap while keeping the same site-balanced structure. These rows are produced directly by `analysis.py`, using the same scoring code as the final method.

Cap c	Maximum Individual Odds	Question-Balanced Brier
0.005	199:1	0.230249
0.010	99:1	0.228757
0.020	49:1	0.228379
0.050	19:1	0.232418

The site-level step is also substantive. HFC sites are hybrid forecasting systems, not ordinary demographic groups. Forecasters within a site may share platform design, information feeds, training, discussion environment, and machine assistance. Treating every forecaster as an independent evidence source can therefore over-count large sites.

10. Step 4 Results and Ablation

Method	Question-Balanced Brier	Question-Day Weighted Brier	Interpretation
Median baseline	0.235399	0.219796	Robust baseline
Original GMO	0.250367	0.233909	Correct but fragile
Capped GMO 0.02	0.232327	0.216881	Bounded evidence only
Site-balanced median	0.234246	0.221594	Site unit correction only
Site-balanced capped GMO 0.02	0.228379	0.211749	Combined method

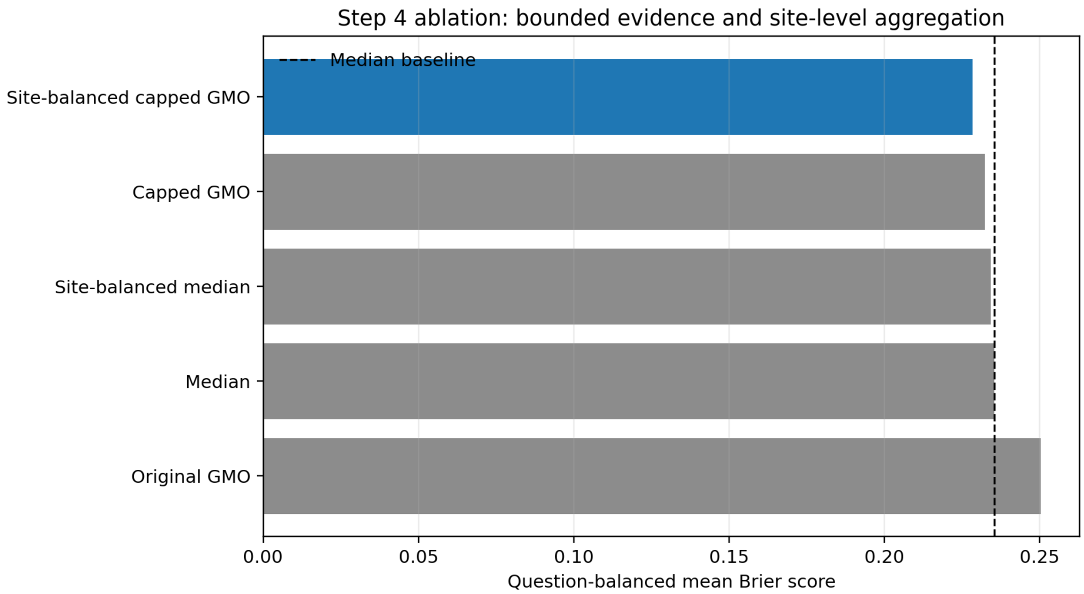

Figure 2: Step 4 ablation

The ablation supports the proposed mechanism. Bounding individual evidence improves over median. Site balancing alone gives a smaller improvement under question-balanced scoring. The combined method performs best, suggesting that individual-level overconfidence and site-level non-exchangeability are distinct failure modes.

By question type, the final method improves the median baseline under question-balanced scoring:

Question Type	Median Baseline	Step 4 Method
Binary	0.189982	0.179451
Nominal	0.478601	0.457898
Ordinal	0.244888	0.243401

The paired bootstrap over questions gives

$$\widehat{R}_{\mathrm{QB}}(\text{median}) - \widehat{R}_{\mathrm{QB}}(\text{Step4}) = 0.0070198,$$

with 95% interval `[-0.0003299, 0.0137224]`. The point estimate is favorable, but the interval slightly crosses zero. I therefore interpret the improvement as promising and mechanistically coherent, not as conclusive out-of-sample evidence.

11. Temporal Validation

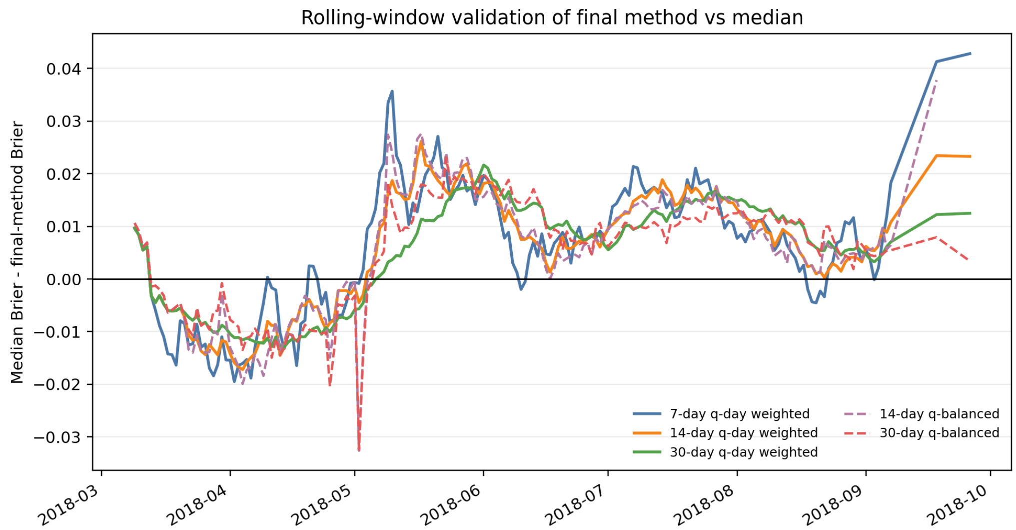

Figure 3: Rolling-window validation

Rolling-window checks ask whether the improvement is concentrated in a short period. They are not external validation, but they are useful temporal diagnostics.

Window	Mean Improvement	Share Positive Windows
7-day daily-weighted	0.005949	0.690608
14-day daily-weighted	0.006045	0.741379
30-day daily-weighted	0.006891	0.797468
45-day daily-weighted	0.007927	0.839161
14-day question-balanced	0.006045	0.741379
30-day question-balanced	0.006891	0.797468
45-day question-balanced	0.007927	0.839161

The fixed method has positive mean improvement across all reported windows and is positive in roughly 69-84% of windows. This reduces the concern that the improvement is driven by a single short episode.

12. Limitations

The sample has only 189 scored questions, and the paired bootstrap interval crosses zero. The result should therefore be validated on another HFC season before being treated as a stable empirical law.

The cap $c = 0.02$ is interpretable, but still a modeling choice. The sensitivity table shows that nearby caps give similar performance, but a stronger design would pre-specify the cap or select it strictly by walk-forward validation.

The site-balanced component has two possible mechanisms: correcting within-site dependence and down-weighting weaker large sites. The data support both as plausible. This is a limitation for causal interpretation, although not necessarily for predictive performance.

The multi-option GMO extension is not unique. I use one-vs-rest odds pooling followed by normalization. Other log-evidence geometries, such as centered log-ratio pooling, may perform differently for nominal and ordinal questions.

Finally, the analysis does not carry forecasts forward. This is consistent with the question-day construction used here, but it limits conclusions about long-horizon staleness and recency decay.

Conclusion

The main statistical finding is that the five baseline methods expose a tradeoff between evidence pooling and robustness. Geometric methods are strong under question-day weighting but vulnerable to per-question tail failures. Median is robust but does not pool evidence on the log-odds scale. Site-balanced bounded evidence pooling resolves this tradeoff by retaining log-odds evidence aggregation while bounding individual influence and correcting the unit of aggregation from forecaster to forecasting site. The resulting method improves both reported risk summaries and has a coherent statistical mechanism, though the size of the improvement should be interpreted conservatively.